This implementation plan provides a step-by-step guide for deploying the FRACA SERVCOM Inventory Management System to a production environment. It covers server requirements (PHP 8.2, MySQL, Nginx/Apache), environment configuration, database setup with seeded admin/manager/staff accounts, asset compilation via Vite, and web server configuration. Post-deployment tasks include caching optimizations, scheduled task configuration for low-stock alerts, security hardening, and verification procedures. A rollback plan and backup strategy are also included to ensure deployment reliability.

INVENTORY MANAGEMENT SYSTEM

IMPLEMENTATION PLAN DOC

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IMPLEMENTATION PLAN (DEPLOYMENT PLAN)

FRACA SERVCOM Inventory Management System

Document Version: 1.0

Date: March 2026

Technology Stack: Laravel 11, PHP 8.2, MySQL, Bootstrap 5, Vanilla JavaScript (ES6+), Vite

1. Introduction

1.1 Purpose

This implementation plan outlines the steps required to deploy the FRACA SERVCOM Inventory Management System to a production environment. It covers server requirements, environment setup, database configuration, asset compilation, and post-deployment verification procedures.

1.2 Deployment Overview

The system is a Laravel 11 web application with session-based authentication and AJAX-driven UI. Deployment involves provisioning a web server, configuring environment variables, installing dependencies, running migrations and seeders, compiling frontend assets with Vite, and configuring the web server.

2. **Server Requirements**

2.1 Minimum Specifications

Component	Requirement
Web Server	Apache 2.4+ or Nginx 1.18+
PHP	8.2 or 8.3
PHP Extensions	BCMath, Ctype, JSON, Mbstring, OpenSSL, PDO, Tokenizer, XML, MySQLi, GD, cURL, Zip
Database	MySQL 8.0+ or MariaDB 10.6+
Web server	Apache 2.4+ or Nginx 1.18+
PHP	8.2 or 8.3
PHP Extensions	BCMath, Ctype, JSON, Mbstring, OpenSSL, PDO, Tokenizer, XML, MySQLi, GD, cURL, Zip
Database	MySQL 8.0+ or MariaDB 10.6+
RAM	2GB minimum (4GB recommended)
Storage	20GB minimum
OS	Ubuntu 22.04 LTS / 24.04 LTS (recommended)

2.2 Required Software

- Composer (PHP dependency manager)
- Node.js 18+ and NPM (for asset compilation)
- Git (for version control)

3. **Pre-Deployment Checklist**

- [ ] Server provisioned with PHP 8.2+ and MySQL
- [ ] Domain name pointed to server IP address
- [ ] SSL certificate installed (Let's Encrypt recommended)
- [ ] Git installed on server
- [ ] Composer installed globally
- [ ] Node.js 18+ and NPM installed
- [ ] Database created with appropriate credentials
- [ ] Firewall configured (ports 80, 443, 22 open)
- [ ] Backup strategy defined

4. **Deployment Steps**

4.1 Server Preparation (Ubuntu Example)

```bash
# Update system
sudo apt update && sudo apt upgrade -y

# Install PHP 8.2 and extensions
sudo apt install -y php8.2 php8.2-cli php8.2-common php8.2-mysql \
php8.2-zip php8.2-gd php8.2-mbstring php8.2-curl php8.2-xml \
php8.2-bcmath php8.2-tokenizer

# Install MySQL
sudo apt install -y mysql-server

# Install Composer
curl -sS https://getcomposer.org/installer | php
sudo mv composer.phar /usr/local/bin/composer

# Install Node.js 18.x
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install -y nodejs
```

```
# Install Nginx (or Apache)
sudo apt install -y nginx

# Install Git
sudo apt install -y git
```

4.2 Application Deployment

```bash
# Clone repository
cd /var/www
git clone https://your-repository-url/fraca-inventory.git
cd fraca-inventory

#Install PHP dependencies
composer install --optimize-autoloader --no-dev

#Install Node dependencies
npm install

# Build frontend assets (production)
npm run build
```

```
# Copy environment file
cp .env.example .env

# Generate application key
php artisan key:generate

# Create storage link
php artisan storage:link
```

4.3 Environment Configuration (.env)

```env
APP_NAME="FRACA SERVCOM Inventory"
APP_ENV=production
APP_DEBUG=false
APP_URL=https://yourdomain.com

# Database Configuration
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=fraca_inventory
DB_USERNAME=inventory_user
```

DB_PASSWORD=secure_password_here

Session Configuration

SESSION_DOMAIN=yourdomain.com

SESSION_SECURE_COOKIE=true

Cache Configuration

CACHE_DRIVER=redis # or file if redis not available

QUEUE_CONNECTION=database

```

4.4 Database Setup

```bash

# Create database and user

sudo mysql -u root -p

```

```sql

CREATE DATABASE fraca_inventory;

CREATE USER 'inventory_user'@'localhost' IDENTIFIED BY 'secure_password_here';

GRANT ALL PRIVILEGES ON fraca_inventory.* TO 'inventory_user'@'localhost';

FLUSH PRIVILEGES;

EXIT;
```

```bash
# Run migrations and seeders
php artisan migrate --force
php artisan db:seed --force
```

Default seeded accounts:

Email	Password	Role
admin@inventory.com	password123	Admin
manager@inventory.com	password123	Manager
staff@inventory.com	password123	Staff

4.5 Web Server Configuration

Nginx Configuration

```nginx
server {
    listen 80;
    server_name yourdomain.com;
    return 301 https://$server_name$request_uri;
```
```

```
}

server {
    listen 443 ssl http2;
    server_name yourdomain.com;
    root /var/www/fraca-inventory/public;

    ssl_certificate /etc/letsencrypt/live/yourdomain.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/yourdomain.com/privkey.pem;

    add_header X-Frame-Options "SAMEORIGIN";
    add_header X-Content-Type-Options "nosniff";
    add_header X-XSS-Protection "1; mode=block";

    index index.php;

    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }

    location ~ \.php$ {
        fastcgi_pass unix:/var/run/php/php8.2-fpm.sock;
        fastcgi_param SCRIPT_FILENAME
$realpath_root$fastcgi_script_name;
        include fastcgi_params;
```

```
    }

    location ~ /\.(?!well-known).* {
        deny all;
    }
}
```

4.6 Set Permissions

```bash
# Set proper ownership
sudo chown -R www-data:www-data /var/www/fraca-inventory
sudo chmod -R 755 /var/www/fraca-inventory/storage
sudo chmod -R 755 /var/www/fraca-inventory/bootstrap/cache

# Restart services
sudo systemctl restart nginx
sudo systemctl restart php8.2-fpm
```

4.7 Configure Scheduled Tasks

```bash
```

Add to crontab

crontab -e

```

Add the following line to run Laravel scheduler every minute:

```

* * * * * cd /var/www/fraca-inventory && php artisan schedule:run >> /dev/null 2>&1

```

**Scheduled tasks include:**

- `GenerateStockAlerts` command: Checks products below reorder level and creates alert records

4.8 Configure Queue Worker (if using database queue)

```bash

# Create supervisor configuration

sudo nano /etc/supervisor/conf.d/laravel-worker.conf

```

```ini

[program:laravel-worker]

process_name=%(program_name)s_%(process_num)02d

```
command=php /var/www/fraca-inventory/artisan queue:work --sleep=3 --tries=3 --max-time=3600

autostart=true

autorestart=true

stopasgroup=true

killasgroup=true

user=www-data

numprocs=2

redirect_stderr=true

stdout_logfile=/var/www/fraca-inventory/storage/logs/worker.log

stopwaitsecs=3600
```

```bash
# Start supervisor

sudo supervisorctl reread

sudo supervisorctl update

sudo supervisorctl start laravel-worker:*
```

5. **Production Optimizations**

5.1 Caching

```bash
php artisan config:cache
php artisan route:cache
php artisan view:cache
php artisan event:cache
```

5.2 Security Hardening

Task	Command/Action
Set secure file permissions	`chmod -R 755 storage bootstrap/cache`
Disable. env viewing	Configure web server to deny access
Enable HTTPS	SSL certificate configured
Set secure session cookies	`SESSION_SECURE_COOKIE=true`
Set secure session cookies	Set secure session cookies
`SESSION_SECURE_COOKIE=true`	
Restrict config caching	`php artisan config:cache`
Set proper .env permissions	`chmod 640 .env`

5.3 Monitoring Setup

```bash
# Install Laravel Telescope (optional, development only)
composer require laravel/telescope --dev
php artisan telescope:install
```

Log locations:
- Application logs: `storage/logs/laravel.log`
- Web server logs: `/var/log/nginx/access.log` and `/var/log/nginx/error.log`
- Worker logs: `storage/logs/worker.log`

6. **Post-Deployment Verification**

6.1 Verification Checklist

- [ ] Application loads at https://yourdomain.com
- [ ] Login works for admin@inventory.com / password123
- [ ] Login works for manager@inventory.com / password123
- [ ] Login works for staff@inventory.com / password123
- [ ] Dashboard displays statistics and widgets
- [ ] Products module: create, edit, view, delete (admin)
- [ ] Products module: delete triggers admin verification modal (manager)
- [ ] Sales module: create sale with dynamic total calculation
- [ ] Purchases module: create purchase with stock increment
- [ ] Reports: generate sales, purchases, stock levels, inventory valuation
- [ ] CSV export downloads correctly from all report tabs
- [ ] Low stock alerts appear on dashboard
- [ ] User management accessible only to Admin
- [ ] No JavaScript console errors
- [ ] Scheduled tasks running (check logs for GenerateStockAlerts)
- [ ] Database backups configured and tested

6.2 Database Connection Test

```bash
php artisan tinker
>>> DB::connection()->getPdo();
>>> DB::table('users')->count();
```

6.3 Schedule Test

```bash
php artisan schedule:list
php artisan schedule:run
```

---

7. **Rollback Procedure**

If deployment issues occur:

```bash
#Restore from previous version
cd /var/www
mv fraca-inventory fraca-inventory-failed
git clone https://your-repository-url/fraca-inventory.git fraca-inventory
cd fraca-inventory

#Copy previous environment and storage
cp ../fraca-inventory-failed/.env ./
cp -r ../fraca-inventory-failed/storage/* storage/

#Restore database from backup
mysql -u inventory_user -p fraca_inventory < /path/to/backup.sql

#Rebuild
composer install --optimize-autoloader --no-dev
npm install
npm run build
php artisan migrate --force
php artisan config:cache
```

```
php artisan route:cache
php artisan view:cache

#Restart services
sudo systemctl restart nginx
sudo systemctl restart php8.2-fpm
```

8. **Maintenance Mode**

To perform maintenance:

```bash
# Enable maintenance mode
php artisan down --retry=60 --message="System undergoing maintenance. Please try again in 1 minute."

# Disable maintenance mode
php artisan up
```

9. **Backup Strategy**

9.1 Database Backup

```bash
# Create backup script
sudo nano /etc/cron.daily/inventory-backup
```

```bash
#!/bin/bash
BACKUP_DIR="/var/backups/fraca-inventory"
DATE=$(date +%Y%m%d_%H%M%S)
mysqldump -u inventory_user -p'secure_password_here' fraca_inventory > $BACKUP_DIR/db_$DATE.sql
# Keep only last 30 days
find $BACKUP_DIR -name "db_*.sql" -mtime +30 -delete
```

```bash
# Make executable
sudo chmod +x /etc/cron.daily/inventory-backup
```

9.2 File Backup

Backup `storage/` directory and `.env` file regularly.

10. **Troubleshooting**

Common Issues

Issue	Solution
Solution	
White screen (500 error)	Check `storage/logs/laravel.log` for details
403 Forbidden	Check `storage/logs/laravel.log` for details
Database connection refused	Verify credentials in .env and MySQL user privileges
CSRF token mismatch	Ensure `SESSION_DOMAIN` matches your domain
Assets not loading	Database connection refused
Scheduler not running	Confirm crontab entry and `php artisan schedule:run` works

11. **Contact & Support**

For deployment issues or questions, contact the development team with relevant log files from `storage/logs/laravel.log`.